

Governing Logistics Under Systemic Disruption: The S³-LOGIC Framework

as a Governance Diagnostic for India's
Multimodal Infrastructure System

B. RAMA RAO AND K. V. RAMAKRISHNA

Correspondence: ramakrishna1161@gmail.com | prof.brrao@andhrauniversity.edu.in



SUSTAINABILITY
(S¹)



SOVEREIGNTY
(S²)



SHOCK-RESILIENCE
(S³)



TOWARDS RESILIENT,
SOVEREIGN AND SUSTAINABLE
LOGISTICS FOR A STRONGER INDIA



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Infrastructure System**

B. Rama Rao and K. V. Ramakrishna

Correspondence: ramakrishna1161@gmail.com

prof.brrao@andhrauniversity.edu.in

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Editorial

India's Logistics Future Will Be Determined by More Than Infrastructure

India's logistics transformation is one of the defining infrastructure and economic governance challenges of its development journey. Dedicated freight corridors, multimodal connectivity initiatives, digital logistics platforms, port modernisation and coordinated infrastructure planning are reshaping how goods move across the country. These investments are essential to improving competitiveness, strengthening domestic supply chains and positioning India more effectively within global trade networks.

Yet infrastructure capacity alone cannot guarantee strategic advantage.

The disruptions of recent years—from the COVID-19 pandemic and geopolitical conflict to maritime insecurity and climate-related shocks—have demonstrated that logistics systems are only as dependable as the institutions responsible for governing them. A corridor may possess modern physical assets and impressive operational capabilities, but its effectiveness can still be undermined by fragmented decision-making, weak crisis coordination, dependence on external digital systems or an inability to sustain environmental gains as freight demand expands.

This is the central question addressed by *Governing Logistics Under Systemic Disruption: The S³-LOGIC Framework as a Governance Diagnostic for India's Multimodal Infrastructure System*. The report shifts attention from infrastructure creation and operational performance towards a less visible, but strategically consequential, dimension of national logistics capability: the quality of governance that sustains the system when conditions become uncertain.

The S³-LOGIC framework brings together three interdependent pillars—**Sustainability, Sovereignty and Shock-Resilience**. Its defining proposition is that these capabilities cannot be treated as interchangeable. Environmental progress cannot compensate for inadequate crisis-response arrangements. Efficient freight movement cannot substitute for institutional control over strategically important logistics data. Nor can digital sophistication, by itself, ensure continuity when infrastructure networks face disruption.

This perspective is particularly relevant to India. The country must expand logistics capacity while simultaneously improving institutional coordination, deepening digitalisation, strengthening domestic strategic autonomy and managing the environmental consequences of growing freight demand. These are not separate policy agendas. They intersect within the same corridors, platforms, agencies and supply chains.

The report's comparative assessment of the Western Dedicated Freight Corridor, the Eastern Dedicated Freight Corridor and the Coastal Maritime Corridor brings this challenge into focus. All three receive a *Fragile* composite governance assessment under the proposed framework, although their individual vulnerability profiles differ. Sovereignty emerges as the weakest of the three assessed pillars across the corridor archetypes, while shock-resilience also presents significant institutional constraints. The findings suggest that physical infrastructure development must be accompanied by stronger arrangements for digital stewardship, coordinated disruption management and adaptive governance.

These results should be understood in the context of the framework's diagnostic purpose. S³-LOGIC is not presented as a replacement for established logistics performance indices. Rather, it seeks to complement them by asking a different question: not simply how efficiently a logistics system performs under prevailing conditions, but whether its governing institutions are equipped to preserve that performance under stress.

That distinction matters for policy. Logistics data governance must address accountability, interoperability, access and control over strategically sensitive information. Disruption preparedness must move beyond reactive coordination towards clearly defined responsibilities, decision pathways and credible recovery arrangements. Sustainability must be assessed not only through infrastructure-level improvements but also through the system-wide consequences of freight growth and modal choices. And these capabilities must be developed across institutional boundaries, including those separating central and state authorities, infrastructure operators, digital platforms and private logistics providers.

The framework also invites a more disciplined approach to measuring governance maturity. By treating the weakest pillar as a binding constraint, it makes deficiencies harder to conceal behind strong performance elsewhere. This non-compensatory approach is valuable because systemic vulnerabilities often emerge precisely where institutions assume that success in one domain will offset weakness in another.

The findings are not a verdict on the achievements of India's logistics reforms. They are a call to protect and extend those achievements. Infrastructure investment creates capacity; sound governance determines how reliably, securely and sustainably that capacity serves the national economy. As logistics networks become more interconnected and digitally mediated, institutional preparedness must be regarded as part of the infrastructure itself.

The contribution of S³-LOGIC lies in offering policymakers, researchers and infrastructure stakeholders a structured way to examine this often-overlooked dimension. Its application to three strategically distinct corridor archetypes provides an initial diagnostic basis for identifying governance gaps and developing more targeted interventions. Further application, refinement and independent validation can help establish its wider utility across India's multimodal logistics ecosystem.

The strategic imperative is clear: India must build not only faster and more connected logistics networks, but also systems capable of withstanding disruption without surrendering sustainability, operational continuity or sovereign control. In an era of recurring uncertainty, resilience is not simply the ability to recover after a crisis. It is the institutional capacity to anticipate vulnerabilities, coordinate action and adapt before a disruption becomes a systemic failure.

The next phase of India's logistics transformation must therefore be measured not only by the infrastructure it creates, but by the governance it builds around it.

Author Declaration on the Use of Artificial Intelligence (AI) Tools

The authors declare that artificial intelligence (AI) tools were used solely as supportive editorial and research-assistance resources during the preparation of this manuscript. AI-assisted language tools were employed to improve grammar and overall linguistic quality without altering the intellectual content, analysis, interpretations, or conclusions of the study. The research concept, study design, analytical framework, data interpretation, findings, policy recommendations, and all scholarly judgments are entirely the original work and responsibility of the authors.

In addition, Jenni was used in a limited capacity to review selected sections for editorial feedback, to identify potential improvements and to assist in verifying consistency of citations and references against the cited sources. The authors assume full responsibility for the accuracy, originality, integrity, and scholarly content of this manuscript, including all analyses, interpretations, citations, and conclusions.

Abstract

Contemporary logistics systems face a compounding governance challenge: operational performance indicators are improving while the institutional foundations sustaining that performance like ecological viability, digital sovereignty, and adaptive resilience under disruption remain structurally underdeveloped. This paper advances the S³-LOGIC Matrices and Index Framework, a governance diagnostic architecture for evaluating multimodal logistics infrastructure in emerging economies operating under recurring geopolitical, environmental, and technological disruption. The framework conceptualises logistics governance across three non-substitutable pillars: Sustainability (S¹), Sovereignty (S²), and Shock-Resilience (S³). The non-substitutability assumption is grounded in logistics systems theory, which recognises that system performance depends on the simultaneous functioning of interdependent governance capabilities rather than on isolated operational strengths. Accordingly, deficiencies in sustainability, sovereignty, or resilience cannot be fully compensated for by superior performance in another dimension because each represents a distinct institutional condition required for the continuity and governance of multimodal logistics systems (Mentzer et al., 2001; North, 1990).

Through Structured Qualitative Comparative Analysis (SQCA) of three strategically distinct Indian freight corridor archetypes (the Western Dedicated Freight Corridor (WDFC), the Eastern Dedicated Freight Corridor (EDFC), and the Coastal Maritime Corridor) the diagnostic reveals a systemic pattern: infrastructure investment is advancing ahead of the governance architecture which is required to protect its outputs under disruption, sustain institutional control over digital operations, and prevent ecological performance from being eroded by demand-induced rebound. All three corridors receive a Fragile composite governance assessment, though their vulnerability profiles are institutionally differentiated. Sovereignty (S²) emerges as the weakest dimension across all corridors, indicating India's logistics digitalisation, (including the expanding implementation of the Unified Logistics Interface Platform (ULIP)), is advancing more rapidly than the institutional governance architecture required to regulate logistics data stewardship, interoperability, platform accountability, and sovereign control over strategic freight information (Department for Promotion of Industry and Internal Trade [DPIIT], 2024). The corridor assessments are further interpreted against the 2024–2025 Red Sea maritime security crisis illustrating how recurrent geopolitical shocks amplify governance vulnerabilities beyond what conventional performance metrics capture. Recent logistics governance research similarly argues that digital transformation without corresponding institutional capacity creates coordination asymmetries and governance vulnerabilities that conventional performance indicators are unable to capture, particularly in rapidly digitalising freight systems (Rodrigue et al., 2024). While the Digital Personal Data Protection Act, 2023 establishes India's national baseline for personal data governance, unresolved institutional gaps remain in logistics data governance, platform control, and algorithmic sovereignty. Seven evidence-based governance prescriptions are derived, aligned to the strategic infrastructure governance imperatives emerging from contemporary geopolitical fragmentation.

Keywords: S³-LOGIC; multimodal infrastructure; institutional resilience; digital sovereignty; adaptive governance; UN Decade of Sustainable Transport

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1. Introduction

1.1 Critical infrastructure governance is emerging as one of the most consequential policy domains of the 2020s. The convergence of geopolitical fragmentation, accelerating digitalisation, and the structural consequences of climate change is exposing institutional vulnerabilities in logistics systems that decades of operational efficiency optimisation have not equipped to manage. Since 2020, the COVID-19 pandemic, the Russia–Ukraine conflict, Red Sea & Suez canal maritime security disruptions, and accelerating extreme weather events have collectively illustrated that logistics systems designed for stable, predictable operating environments are systemically exposed once those conditions no longer hold.

1.2 Freight logistics is not simply an economic sector; it is the connective architecture of national productive capacity. When logistics governance is inadequate, the consequences cascade across food security, industrial supply continuity, export competitiveness, and fiscal stability. For emerging economies undertaking rapid logistics transformation where infrastructure investment is accelerating, digitalisation is deepening, and disruption frequency is increasing simultaneously the governance challenge is qualitatively different from that facing established logistics economies. Unlike mature logistics systems, emerging economies must simultaneously expand infrastructure capacity, establish new governance institutions, and coordinate multiple public and private actors under resource and institutional constraints, creating governance complexities that extend beyond conventional infrastructure development (Vining et al., 2023). This governance architecture lag reflects a well-recognised implementation challenge in critical infrastructure governance, where infrastructure investment frequently advances faster than the institutional arrangements needed for cross-sectoral coordination, regulatory integration, and adaptive decision-making across multiple levels of government (Scholten et al., 2024; Peters, 2018). The question is not how to maintain governance of a stable system, but how to build governance architecture in a system that is transforming under disruption pressure.

1.3 India exemplifies this challenge. Through the National Logistics Policy (2022) the PM GatiShakti National Master Plan (2021) and sustained Dedicated Freight Corridor investment, the Government of India has achieved logistics costs of roughly 7.97% of GDP towards globally competitive levels. While these initiatives have substantially strengthened physical connectivity and logistics integration, governance capacity varies across Indian states owing to the country's federal institutional structure and uneven implementation of logistics reforms. (Rodríguez-Pose & Gill, 2003). This paper argues that the institutional governance architecture required to protect, sustain, and extend those infrastructure gains under contemporary disruption conditions is not advancing at a comparable pace, and that this asymmetry constitutes a strategic vulnerability that conventional logistics performance metrics cannot detect.

1.4 This paper makes three substantive contributions. First, it advances the S³-LOGIC Matrices and Index Framework: an integrative governance diagnostic architecture that simultaneously assesses ecological viability (S¹ Sustainability), institutional agency and digital control (S² Sovereignty), and adaptive operational continuity (S³ Shock-Resilience). The framework conceptualises Sustainability (S¹), Sovereignty (S²), and Shock-Resilience (S³) as jointly necessary and non-substitutable governance conditions because multimodal logistics systems function as integrated socio-technical networks in which institutional deficiencies in one governance capability cannot be offset by strengths in another, thereby making each pillar a necessary condition for system-wide governance effectiveness (Mentzer et al., 2001; Christopher & Peck, 2004). Second, it applies S³-LOGIC empirically across three

structurally differentiated Indian freight corridor archetypes, generating institutionally specific governance-vulnerability profiles. Third, it derives governance prescriptions aligned to both the UN Decade of Sustainable Transport and the strategic imperatives shaping global infrastructure governance. Although the 2023 World Bank Logistics Performance Index (LPI) incorporates indicators relating to digitalisation, supply chain resilience, and cybersecurity([Arvis et al., 2023](#)), its evaluative focus remains operational and performance-oriented rather than assessing institutional authority, governance maturity, sovereign control of logistics data, inter-agency coordination, or adaptive decision architectures under geopolitical disruption. These institutional attributes constitute the analytical focus of S³-LOGIC.

1.5 The necessity for a new governance diagnostic arises not because existing logistics frameworks are ineffective, but because they were designed to answer fundamentally different policy questions. Contemporary benchmarking instruments such as the Logistics Performance Index (LPI), Green Logistics Performance Index (GLPI), Liner Shipping Connectivity Index (LSCI), OECD resilience frameworks, and UNCTAD port assessments provide valuable insights within their respective domains. However, these instruments predominantly evaluate operational performance, connectivity, sustainability outcomes, or resilience capacities in isolation. None simultaneously examines whether the institutional architecture governing a logistics system possesses the capacity to sustain environmental performance, retain sovereign control over critical logistics data, and maintain operational continuity during systemic disruption. As geopolitical fragmentation, digital platform dependence, and climate uncertainty increasingly converge, governance itself becomes a strategic infrastructure asset rather than merely an administrative function. It is precisely this institutional diagnostic gap that the S³-LOGIC framework addresses by treating Sustainability, Sovereignty, and Shock-Resilience as jointly necessary and institutionally inseparable conditions for future-ready multimodal logistics governance.

2. Theoretical and Empirical Context

2.1 The governance of critical infrastructure has evolved from a technical-operational concern into a strategic institutional challenge and governance in complex systems requires collaborative institutions managing authority across multiple actors. In logistics infrastructure this challenge is acute, since governance authority is typically distributed across transport ministries, regulatory agencies, commercial operators, and subnational governments : what [Ansell & Gash, \(2007\)](#) characterise as collaborative governance problems in which no single actor holds sufficient authority or information to govern effectively alone. Institutional theory offers a foundational explanation for why governance architecture fails to develop in parallel with operational capability.

2.2 Drawing on [\(DiMaggio & Powell, \(1983\)\)](#)of institutional isomorphism, three mechanisms explain the governance-architecture lag this paper diagnoses across all three corridor archetypes. Coercive isomorphism (driven by central-government mandates such as PM GatiShakti and the National Logistics Policy 2022) channels institutional energy into measurable infrastructure investment while leaving the less visible architecture of data sovereignty, disruption protocols, and MSME inclusion unaddressed. Mimetic isomorphism leads governance designers to emulate established instruments (principally the World Bank LPI) that were never calibrated to measure institutional control or resilience architecture, reproducing the same measurement blind spots across successive policy cycles. Normative isomorphism, transmitted through professional logistics networks and consultancy ecosystems, reinforces operational efficiency as the dominant legitimacy criterion, further marginalising sovereignty and resilience governance.

2.3 Institutional-pillar framework adds precision and the provisions S² and S³ find absent are regulative institutions like legally documented protocols requiring deliberate legislative construction, rather than institutions that emerge organically from practice. This distinction is consequential, because regulative gaps cannot close incrementally; they require explicit policy intervention, which is what this paper's seven prescriptions provide. Although institutional theory explains why governance architectures often evolve more slowly than physical infrastructure, the consequences of this lag are particularly acute in strategically important logistics systems.

2.4 Institutional voids rarely emerge because regulatory attention is naturally drawn towards visible physical assets than governance mechanisms (including data stewardship, crisis decision authority, inter-agency coordination protocols, and digital sovereignty) and develop incrementally or remain fragmented. Over time, these governance gaps become reinforced through path dependency, whereby early institutional choices constrain future policy options and increase the cost of corrective intervention. Consequently, governance deficits become progressively more difficult to reverse as logistics platforms mature, digital ecosystems consolidate, and operational dependence on existing institutional arrangements deepens. This institutional perspective explains why governance reform must precede, rather than follow, large-scale logistics digitalisation if long-term strategic autonomy and systemic resilience are to be preserved.

2.5 The digitalisation of logistics infrastructure has created a new class of governance risk: the progressive cession of operational control to commercial platforms. Digital sovereignty [\(Floridi, 2020\)](#); [\(Zuboff, 2019\)](#) converges on the insight that platform dependency risks are path-dependent and progressive as once created, they are difficult to reverse. Applied to logistics, this literature implies that data governance, algorithmic accountability, and platform-concentration oversight are institutional conditions rather than technical settings, a distinction this paper's S² pillar

operationalises directly. Beyond digital sovereignty, logistics systems should be understood as adaptive governance systems rather than static transport networks. While physical infrastructure provides the capacity to move freight, governance architecture determines how effectively that capacity can be reconfigured under conditions of uncertainty.

2.6 Dynamic capabilities theory suggests that resilient logistics systems require three interdependent institutional capabilities: the ability to sense emerging disruptions before they cascade across the network, the authority to seize and coordinate rapid responses through predefined governance mechanisms, and the flexibility to reconfigure operational arrangements as disruption profiles evolve. These capabilities cannot be improvised during a crisis and they must be institutionalised through documented protocols, interoperable data architectures, and clearly defined decision authority. Accordingly, S³-LOGIC evaluates not merely the presence of infrastructure, but the governance capability that enables infrastructure to remain functional, adaptive, and strategically effective during periods of systemic disruption.

2.7 Resilience governance has emerged as a distinct subfield addressing institutional capacity for disruption response. [Wıldavsky, \(1988\)](#) foundational distinction between anticipatory and resilient governance strategies, extended by [Folke et al., \(2010\)](#) and [Boin et al., \(2017\)](#), establishes that governance systems capable of absorbing structural shocks and enabling institutional recovery differ qualitatively from systems optimised only for stable-state efficiency. The 2024–2025 Red Sea crisis, alongside COVID-19 and the Russia–Ukraine conflict, reinforces this literature empirically: recent Suez Canal disruptions affecting Indian trade demonstrate that external geopolitical shocks can rapidly propagate into domestic freight systems even where physical infrastructure performance is unaffected.

2.8 A growing body of work on resilience governance in emerging-economy logistics contexts finds consistent departures from advanced-economy assumptions of stable institutional baselines, yet the institutional-governance dimension of supply chain resilience remains comparatively under-examined; Chopra and Sodhi (2014) emphasise operational risk mitigation rather than the institutional architecture within which that mitigation must occur ([Chopra & Sodhi, 2014](#)). The 2024–2025 Red Sea maritime security crisis illustrates this distinction with particular clarity. India's freight infrastructure remained physically intact, yet shipping routes were diverted around the Cape of Good Hope, transit times lengthened, freight rates increased, inventory planning became more uncertain, and logistics operators were compelled to revise sourcing and routing strategies. These outcomes were driven not by infrastructure failure, but by governance and coordination challenges extending across multiple jurisdictions and commercial actors. The episode demonstrates that contemporary logistics resilience increasingly depends upon institutional preparedness, decision-making authority, data visibility, and cross-sector coordination rather than infrastructure capacity alone. In this context, geopolitical disruption should be understood as a governance stress test that reveals institutional vulnerabilities otherwise concealed during periods of operational normalcy, reinforcing the rationale for evaluating governance maturity alongside conventional logistics performance indicators.

2.9 India's logistics governance challenge has generated a substantial body of policy-oriented research indicating that, despite important improvements achieved by several states under India's competitive federal framework, nationally coordinated governance arrangements for multimodal freight systems remain institutionally uneven, particularly in areas requiring inter-state coordination, digital interoperability, and

common regulatory standards (DPIIT & NCAER, 2023; NITI Aayog, 2022). India's logistics costs at 7.97% of GDP; reflects methodological refinement alongside genuine efficiency gains. PM GatiShakti represents the most ambitious attempt at governance coordination undertaken in Indian logistics, aligning infrastructure planning across sixteen ministries; PM GatiShakti and the Unified Logistics Interface Platform (ULIP) have substantially strengthened infrastructure planning and digital integration; however, neither currently establishes comprehensive governance provisions for operational logistics data stewardship, platform accountability, algorithmic transparency, or sovereign oversight of strategically significant freight information, thereby reinforcing the governance gap addressed by the proposed National Logistics Data Sovereignty Act (Department for Promotion of Industry and Internal Trade [DPIIT], 2024).

2.10 The coastal maritime governance gap is particularly well documented: Despite sustained investment under the Sagarmala Programme, coastal shipping continues to account for only a modest share of India's domestic freight movement, with recent policy assessments attributing this to persistent governance, regulatory, and multimodal connectivity constraints rather than infrastructure deficiencies alone (Ministry of Ports, Shipping and Waterways, 2024). Sagarmala investment has addressed port infrastructure, but the regulatory and institutional barriers to coastal modal activation, cabotage complexity, container equipment availability, and port tariff asymmetries remain unaddressed as governance problems.

2.11 The World Bank Logistics Performance Index, first published in 2007, remains the dominant global benchmark for national logistics assessment. Despite its institutional authority, it assesses six operational dimensions (customs efficiency, infrastructure quality, ease of arranging shipments, service quality, tracking and tracing, and timeliness) but contains no governance dimension, even though governance quality exerts the strongest effect on logistics performance in cross-country studies. Arvis et al. (2016) themselves acknowledge that the LPI captures current performance rather than the governance conditions determining whether that performance is sustainable or resilient. The OECD's ITF Transport Outlook similarly provides comprehensive modal performance benchmarks without governance-diagnostic instruments applicable to specific infrastructure systems. Nor is the LPI alone in this respect.

2.12 The Green Logistics Performance Index anticipates S³-LOGIC's S¹ pillar by asking whether logistics efficiency and ecological performance move together, but it measures outcome correlation rather than the institutional mechanisms (demand-moderation instruments, absolute-load monitoring, energy-transition governance) that would sustain ecological performance under demand growth. The Liner Shipping Connectivity Index offers the methodologically most mature parallel to S³-LOGIC's ambitions, built on nearly two decades of objective connectivity data, but it cannot distinguish a well-connected port under national institutional control from one whose connectivity rests entirely on foreign terminal-operator data governance—precisely the distinction the S² pillar is designed to make analytically visible. The OECD Supply Chain Resilience Framework (2025) offers the most theoretically sophisticated existing treatment of disruption governance, but its indicators operate at national macroeconomic scale and cannot assess whether a specific freight corridor possesses a documented rerouting protocol or pre-authorised crisis authority; S³-LOGIC operationalises the OECD's own vocabulary (agility, adaptability, alignment) at the corridor-operational level where governance investment decisions are actually made.

2.13 Finally, the UNCTAD Port Performance Scorecard (2024) operates at the most granular level, individual port management, with a self-reported, operational-financial scope that cannot observe data-sovereignty provisions, algorithmic transparency, or inter-port rerouting governance. The collective limitation of these instruments is one of diagnostic scope that they measure static performance outcomes effectively but are structurally unable to assess the institutional governance capacity that produces, or fails to protect, those outcomes. S³-LOGIC is designed to fill precisely this diagnostic gap, not to supersede any of these instruments within their own domains. This positioning complements, rather than replaces, existing international assessment frameworks. Whereas the OECD Infrastructure Governance Indicators (2023) evaluate national governance capacity and the Trade Corridor Efficiency Index (2025) primarily measures corridor operational performance, S³-LOGIC uniquely diagnoses corridor-level institutional maturity by integrating sustainability, sovereignty, and adaptive resilience within a single governance framework.

2.14 Importantly, S³-LOGIC should not be interpreted as an alternative to existing logistics assessment frameworks, but as a complementary governance diagnostic that extends their analytical scope. Performance-oriented instruments remain indispensable for measuring operational efficiency, infrastructure quality, connectivity, sustainability outcomes, and trade competitiveness. However, high operational performance does not necessarily imply governance maturity. A corridor may demonstrate excellent freight throughput, modern infrastructure, and improved logistics efficiency while simultaneously remaining vulnerable to institutional fragmentation, digital platform dependency, inadequate crisis-governance mechanisms, or weak sovereign control over strategically significant logistics data. By explicitly separating governance capacity from operational performance, S³-LOGIC enables policymakers to identify vulnerabilities that conventional performance indicators may inadvertently conceal. This distinction is particularly significant for emerging economies where infrastructure expansion, digital transformation, and geopolitical uncertainty are unfolding simultaneously, requiring governance capability to evolve at least as rapidly as physical infrastructure itself. Table 1 traces the chronological evolution of governance theory and demonstrates how successive paradigms collectively converge into the proposed S³-LOGIC governance diagnostic framework. It positions S³-LOGIC as an integrative extension rather than a departure from established governance scholarship.

Table 1. Evolution of Governance Theory toward S³-LOGIC Governance

Era	Dominant Paradigm	Key Scholars	Core Assumption	Relevance to S ³ -LOGIC
Pre-1990	Command-and-Control Public Administration	Weber (1922); Taylor (1911)	Top-down governance suffices for complex systems.	Explains absence of multi-actor coordination architecture in legacy Indian freight governance.
1990–2000	New Public Management	Hood (1991); Osborne & Gaebler (1993)	Market mechanisms and performance metrics drive governance quality.	Origin of LPI-style performance-index governance templates dominating logistics policy design.
2000–2010	Collaborative / Network Governance	Ansell & Gash (2008); Provan & Kenis (2008)	No single actor holds sufficient authority; governance requires multi-actor coordination.	Theoretical foundation for S ³ -LOGIC's diagnosis of multi-stakeholder governance failure (EDFC).
2010–2020	Adaptive and Resilient Governance	(Folke et al., 2010) ; Boin et al. (2016)	Governance must absorb structural change and enable institutional recovery from disruption.	Theoretical grounding for the S ³ pillar's agility, adaptability, and alignment sub-dimensions.
2020–present	Digital and Platform Governance	Srnicek (2017); Floridi (2020); Flyverbom et al. (2019)	Digital platforms are governance actors; sovereignty over data and algorithms is institutional, not technical.	Direct theoretical basis for the S ² pillar and Prescriptions G1, G2, G6, G7.
S ³ -LOGIC (2026)	Integrated Tri-Governance Diagnosis		Sustainability, sovereignty, and resilience are jointly necessary, non-substitutable governance conditions.	Central contribution of this paper, synthesising prior paradigms into a unified corridor-level diagnostic.

Source: Developed by the authors (2026), synthesised from Weber (1922), Taylor (1911), Hood (1991), Osborne and Gaebler (1993), Ansell and Gash (2008), Provan and Kenis (2008), Folke et al. (2010), Boin et al. (2016), Srnicek (2017), Floridi (2020), and Flyverbom et al. (2019).

Note: Historical periods are indicative rather than absolute and represent dominant theoretical shifts in governance scholarship. The table is interpretive and conceptual, illustrating how successive governance paradigms contribute to the intellectual foundations of S³-LOGIC rather than implying linear theoretical replacement.

2.15 Table 1 demonstrates a progressive expansion in governance thinking from hierarchical administrative control towards integrated, adaptive, and digitally enabled governance architectures. Early paradigms prioritised administrative authority and efficiency, whereas subsequent frameworks increasingly recognised multi-actor coordination, institutional adaptability, and platform governance as essential responses to growing systemic complexity. This theoretical progression reveals that logistics governance has evolved beyond operational management towards managing institutional resilience, digital sovereignty, and cross-sector coordination under persistent disruption. S³-LOGIC synthesises these cumulative intellectual developments by integrating sustainability, sovereignty, and shock-resilience into a unified corridor-level governance diagnostic. Rather than replacing preceding paradigms, the framework consolidates their complementary strengths while addressing governance dimensions overlooked by conventional performance-oriented approaches. The evolution presented in the table therefore provides the theoretical justification for treating governance capability as a strategic infrastructure asset and supports the study's central proposition that institutional maturity, rather than operational efficiency alone, determines the long-term robustness and adaptive capacity of multimodal logistics systems.

2.16 Table 2 consolidates the principal theoretical foundations underpinning the S³-LOGIC framework by linking established governance theories with their institutional manifestations in Indian freight governance and corresponding policy prescriptions.

Table 2. Theoretical Foundations of the S³-LOGIC Pillars

Theoretical Source	Core Mechanism	Manifestation in Indian Freight Governance	Pillar(s) Diagnosed	Prescription Link (Para 9)
Coercive isomorphism — DiMaggio & Powell (1983)	Central-government mandates drive conformity to dominant institutional templates.	PM GatiShakti and the National Logistics Policy (2022) have strengthened nationally coordinated infrastructure planning; however, the institutional maturity of governance mechanisms for digital logistics, data stewardship, and disruption management continues to vary across states, reflecting India's federal implementation landscape (NITI Aayog, 2022; DPIIT & NCAER, 2023).	S ¹ , S ² , S ³	G1, G3

Theoretical Source	Core Mechanism	Manifestation in Indian Freight Governance	Pillar(s) Diagnosed	Prescription Link (Para 9)
Mimetic isomorphism — DiMaggio & Powell (1983)	Organisations emulate successful peers under ambiguity, adopting dominant templates without scrutiny.	LPI-derived templates exclude sovereignty and resilience dimensions; NLP benchmarks use LPI-rank improvement as a governance target.	S ² , S ³	G6, G7
Normative isomorphism — DiMaggio & Powell (1983)	Professional networks transmit shared norms privileging efficiency as the dominant legitimacy criterion.	Logistics consultancy ecosystem prioritises speed and cost; data, disruption, and MSME governance treated as secondary.	S ² , S ³	G2, G3
Regulative institutions — Scott (2008); North (1990)	Formal, legally documented protocols with coercive enforcement, requiring deliberate legislative construction.	No documented national data-sovereignty framework; absence reflects a regulative institutional void, not organic gradual formation.	S ² , S ³	G1, G3, G6
Logistics systems theory — Mentzer et al. (2001); Lambert & Cooper (2000)	Integrated logistics performance depends on coordinated institutional as well as operational architecture.	Modal efficiency gains (WDFC, EDFC) are not matched by coordinated data or crisis-governance architecture.	S ¹ , S ³	G3, G5
Dynamic capabilities theory — Teece et al. (1997); Beske (2012)	Sensing, seizing, and reconfiguring capacities determine institutional resilience under change.	No institutionalised sensing (monitoring), seizing (rapid activation), or reconfiguring (rerouting) capability documented in any corridor.	S ³	G3

Theoretical Source	Core Mechanism	Manifestation in Indian Freight Governance	Pillar(s) Diagnosed	Prescription Link (Para 9)
Digital platform governance — Srnicek (2017); Evangelista et al. (2013)	Platform dependency in logistics creates a governance condition, not merely a market-structure observation.	ULIP, DFC booking platforms, and port concession systems have significantly strengthened logistics digital integration; however, governance mechanisms for trusted operational data sharing, platform accountability, data stewardship, and sector-specific digital sovereignty remain underdeveloped despite emerging international Trusted Data Space (TDS) and International Data Spaces (IDS) governance models (International Data Spaces Association [IDSA], 2024; Catena-X Automotive Network, 2023).	S ²	G1, G6, G7
Resource-advantage theory — Hunt & Davis (2008)	Governance architecture is itself a rare, path-dependent, socially complex resource, unlike physical infrastructure.	All three corridors receive a Fragile composite despite substantial infrastructure investment, confirming governance is not a by-product of investment.	S ¹ , S ² , S ³	All seven

Source: Developed by the authors (2026), synthesised from DiMaggio and Powell (1983), Scott (2008), North (1990), Mentzer et al. (2001), Lambert and Cooper (2000), Teece et al. (1997), Beske (2012), Srnicek (2017), Evangelista et al. (2013), and Hunt and Davis (2008).

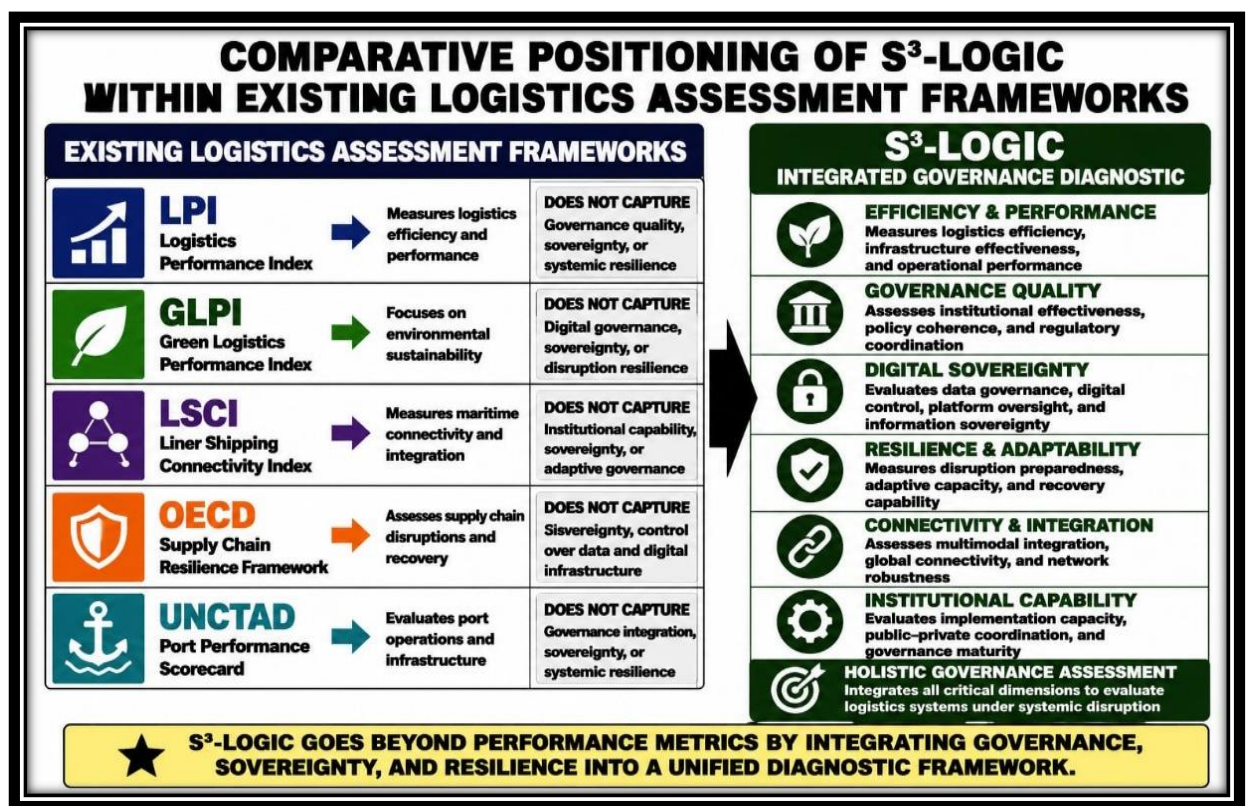
Note: The table operationalises established governance theories within the Indian multimodal freight context by mapping each theoretical mechanism to observed governance conditions, the affected S³-LOGIC pillar(s), and the corresponding governance prescriptions proposed in this study.

2.17 Table 2 demonstrates that the S³-LOGIC framework is theoretically cumulative rather than conceptually novel in isolation. Institutional, logistics, resilience, digital governance, and resource-based theories collectively explain the governance deficiencies observed across Indian freight corridors. Isomorphic pressures account for governance convergence, institutional theory explains regulatory voids, logistics systems and dynamic capabilities emphasise coordinated resilience, while digital platform governance highlights emerging sovereignty challenges. Resource-advantage

theory further establishes governance capability as a strategic institutional asset rather than an automatic outcome of infrastructure investment. Collectively, these theoretical linkages provide a coherent explanatory foundation for the three S³-LOGIC pillars and justify the governance prescriptions advanced in the study, thereby strengthening the framework's analytical validity and policy relevance.

2.18 Figure 1 positions the S³-LOGIC Framework relative to established logistics assessment frameworks. It highlights that S³-LOGIC complements rather than replaces existing indices by extending evaluation from operational performance to governance capability under systemic disruption.

Figure 1. Comparative Positioning of S³-LOGIC within Existing Logistics Assessment Frameworks



Source: Developed by the Authors (2026), based on synthesis of the World Bank LPI, Green Logistics Performance Index (GLPI), Liner Shipping Connectivity Index (LSCI), OECD resilience frameworks, and UNCTAD port performance frameworks.

Note: Existing frameworks primarily assess operational efficiency, sustainability, connectivity, or resilience within their respective domains. S³-LOGIC extends these perspectives by integrating governance capability, institutional maturity, digital sovereignty, and shock resilience into a unified diagnostic architecture.

2.19 Figure 1 demonstrates the principal theoretical contribution of S³-LOGIC. Rather than competing with established logistics indices, the framework complements them by evaluating institutional governance capabilities that conventional performance-oriented assessments do not explicitly measure. This positioning establishes S³-LOGIC as a governance diagnostic rather than another operational performance index.

3. Research Gap and Theoretical Positioning:

3.1 Three interrelated diagnostic gaps motivate this study. A measurement gap: no existing instrument assesses governance conditions (institutional capacity, coordination architecture, digital sovereignty, resilience protocols) as distinct from performance outcomes. A contextual gap: no existing framework is calibrated for the simultaneous growth, inclusion, and transition pressures facing rapidly transforming emerging-economy logistics systems under compound disruption. And a temporal gap: no existing framework evaluates governance trajectories under conditions of recurring disruption rather than assumed operational stability. S³-LOGIC is propagated as a governance diagnostic architecture, integrating multi-level and collaborative governance theory into a single tri-pillar assessment instrument. Its defining theoretical commitment is non-substitutability as no governance dimension can compensate for institutional deficit in another. The minimum-condition rule follows configurational assessment principles in which the absence of a necessary governance capability constrains overall system performance irrespective of strengths elsewhere. This approach is consistent with composite indicator methodology for institutional assessment, where non-compensatory aggregation is appropriate when constituent dimensions represent jointly necessary governance conditions rather than substitutable performance attributes (Nardo et al., 2008; Ray, 2014). This distinguishes S³-LOGIC from additive composite indices such as the LPI, whose weighted averaging allows strength in one dimension to mask fragility in another a masking effect that Stević et al. (2024) document empirically in LPI data. The non-substitutability rule instead implements the theoretical proposition that resilient, sustainable, and sovereign outcomes are jointly necessary conditions and is institutionally grounded in (North, 1990) account of formal institutions: because the provisions S² and S³ diagnose are regulative rather than normative or cognitive and their absence cannot be partially offset by goodwill or awareness.

3.2 Research Objectives, Hypothesis, and Causal Mechanism

3.2.1 The primary objective is to develop, apply, and partially validate a governance diagnostic framework capable of assessing whether India's multimodal freight corridors possess the institutional architecture required to deliver sustainable, sovereign, and resilient outcomes under contemporary disruption conditions. Three sub-objectives operationalise this aim: (i) identify and characterise governance-architecture deficits that established performance frameworks are not designed to diagnose; (ii) Apply the S³-LOGIC instrument across three structurally differentiated Indian freight corridor archetypes as analytically representative governance contexts, thereby generating governance-vulnerability profiles differentiated by corridor function, social consequence, and disruption profile while demonstrating the framework's applicability across diverse multimodal logistics settings rather than as an exhaustive assessment of India's logistics system and (iii) Derive evidence-grounded governance prescriptions aligned to the 2030 Agenda for Sustainable Development and to contemporary geopolitical imperatives.

3.2.2 **H1 (Central Hypothesis):** Freight corridor archetypes that receive lower composite S³-LOGIC governance scores reveal governance-vulnerability profiles characterised by **(a)** absent or undocumented institutional disruption-response protocols, **(b)** structural exclusion of MSME operators from interoperable digital logistics ecosystems, and **(c)** the absence of sector-specific

governance mechanisms for operational logistics data stewardship, trusted data sharing, and sovereign oversight within digital logistics platforms, including the Unified Logistics Interface Platform (ULIP). H1 does not assert that S³-LOGIC is superior to existing frameworks but it asserts that S³-LOGIC identifies governance dimensions that existing frameworks do not capture, and that variation in these dimensions is associated with variation in operational vulnerability—a relationship assessed through SQCA and expert practitioner validation, and interpreted as an illustrative directional pattern rather than as inferential statistical evidence, given the study's $n = 3$ corridor design.

3.2.3 The diagnostic claims rest on a theoretically grounded, five-step causal mechanism rather than an assumed correlation. Consistent with the paper's critical realist stance (Bhaskar) governance structures are understood as real institutional mechanisms producing observable operational tendencies through context-dependent pathways and not as invariant causal laws. First, governance architecture shapes institutional capability: where regulative institutions (statutory data-sovereignty frameworks, pre-authorised crisis authorities, documented rerouting protocols) exist, institutional actors possess the legal authority and coordination mandate to govern effectively and where they are absent, across all three corridors. Second, institutional capability shapes logistics performance, since the consistency and resilience of freight movement depend on governance of modal coordination, pricing transparency, and booking access and not physical infrastructure alone (Mentzer et al., 2001). Third, performance shapes resilience: systems with institutional capability for rapid rerouting and cascade isolation recover from disruption more quickly, as the COVID-19 and Red Sea evidence discussed later illustrates. Fourth, resilience shapes sustainability, since governance-fragile systems revert to carbon-intensive road fallback under disruption, eroding cumulative sustainability gains (Folke et al., 2010). Fifth, these pathways jointly determine corridor outcomes, with the minimum-condition rule ensuring that the weakest governance pathway, not the average, determines the composite governance band, consistent with configurational logic in Qualitative Comparative Analysis, where the absence of a necessary condition constrains overall system performance irrespective of strengths elsewhere (Ragin, 2008). This mechanism is consistent with, but not proven by, the strong directional association between composite governance and vulnerability scores reported in Table 5 (the small- N design precludes causal inference, and extending the empirical base to $n \geq 10$ corridors is the appropriate next step for testing the mechanism inferentially).

3.2.4 The principal innovation of S³-LOGIC lies not in introducing an additional performance index, but in reconceptualising logistics governance as a system of non-substitutable institutional conditions. Existing assessment frameworks generally aggregate indicators through weighted averaging, permitting strong performance in one dimension to offset weaknesses in another. Such an approach is appropriate for operational benchmarking but is less suitable for governance diagnostics, where the absence of a critical institutional capability may compromise the entire system regardless of strengths elsewhere. S³-LOGIC therefore adopts a minimum-condition logic, recognising Sustainability (S¹), Sovereignty (S²), and Shock-Resilience (S³) as jointly necessary governance pillars whose effectiveness depends upon their simultaneous institutional maturity. This conceptual shift enables the framework to identify hidden governance vulnerabilities that remain obscured within conventional composite indices and provides policymakers with a more

strategically actionable basis for prioritising governance reform in complex multimodal logistics systems operating under persistent geopolitical, technological, and environmental uncertainty.

4. Research Methodology

4.1 The study adopts a critical realist research philosophy, treating governance structures as real institutional mechanisms whose effects are context-dependent rather than deterministic. The three governance pillars are treated as necessary institutional conditions, since sustained governance performance requires simultaneous ecological sustainability (S¹), institutional sovereignty (S²), and adaptive resilience (S³); deficiency in any pillar constitutes institutional fragility irrespective of strengths elsewhere. Consistent with configurational methodology (Schneider & Wagemann, 2012), the analysis recognises equifinality and causal asymmetry that governance vulnerability may arise through multiple institutional configurations, and the conditions associated with governance maturity are not assumed to be the simple inverse of those producing fragility.

4.2 The primary analytical method is Structured Qualitative Comparative Analysis (SQCA): a pre-defined, theoretically grounded instrument applied consistently across cases to generate comparable diagnostic profiles, distinguished from inductive thematic analysis by its use of a pre-specified framework and from quantitative comparative analysis by its interpretive rather than statistical logic (building on Ragin (2008)). The diagnostic follows a transparent four-step process for each pillar within each corridor: structured review of secondary data against the pillar's four sub-dimensions; qualitative assessment of trajectory (present-and-improving, present-and-stalled, or absent); band assignment using the explicit decision rules in Table 4, applied consistently across all corridors; and documentation of the evidence-to-score reasoning chain, creating an auditable governance-assessment record consistent with the OECD Recommendation on the Governance of Infrastructure (2023).

4.3 Scores are directional diagnostic assessments and not cardinal measurements: a pillar score of 1.8 signals that governance provisions are predominantly absent and the trajectory unclear and not that the corridor has achieved 36% of full governance capacity. The minimum-condition rule (whereby the composite band is determined by the weakest governance pillar) operationalises the framework's non-substitutability principle by treating sustainability, sovereignty, and resilience as jointly necessary institutional conditions. While this conservative aggregation may not fully capture corridor-specific governance strengths arising from subnational implementation differences, it is intentionally adopted to identify the most critical governance constraint requiring policy intervention, consistent with the logic of necessary conditions in Qualitative Comparative Analysis (Ragin, 2008; Schneider & Wagemann, 2012).

4.4 Five institutionally authoritative sources provide the empirical foundation[NITI Aayog (2023), World Bank LPI(2023), DPIIT–NCAER (2023) Logistics Cost Assessment, Ministry of Railways (2023) Annual Report; and PM GatiShakti (2024) Progress Reports], supplemented by a systematic literature review conducted in accordance with the PRISMA Statement to ensure identification, screening, and inclusion of relevant evidence Expert validation was conducted through 19 semi-structured interviews across ten professional categories in March 2026, spanning logistics practitioners, freight corridor operators, policymakers, academic specialists, and military logistics professionals, confirming that band assignments derived from secondary data were consistent with practitioner experience and surfacing governance dimensions not captured in published data. All respondents provided informed consent, and all data is fully anonymised, with institutional role attributions used in place of individual identification.

4.5 With $n = 3$ corridor observations, the quantitative findings presented are directional and illustrative rather than statistically generalisable. The Pearson correlation between composite governance and vulnerability scores ($r = -0.9993$) is reported as an illustrative pattern consistent with H1, not as inferential evidence, and the paper is appropriately characterised as governance-oriented comparative diagnostic research. The small sample size precludes reliable estimation of population-level relationships and primarily serves to demonstrate the internal consistency of the proposed governance diagnostic framework (Yin, 2018; Schneider & Wagemann, 2012). Future research extending the empirical base to $n \geq 10$ corridors across multiple emerging-economy contexts would enable formal inferential testing. Table 3 presents the operationalisation of the study variables by linking each governance construct with its conceptual definition, empirical evidence proxies, and corresponding data sources used in the S³-LOGIC diagnostic framework.

Table 3. Variable Operationalisation - Independent and Dependent Variables

Variable	Conceptual Definition	Governance Evidence Proxy	Source
S ¹ Sustainability (IV ₁)	Degree to which governance operates within ecological constraints and addresses rebound risk.	Modal efficiency trajectory; environmental load monitoring; demand-moderation governance.	NITI Aayog (2023); Ministry of Railways (2023)
S ² Sovereignty (IV ₂)	Degree to which national institutions retain governance agency over logistics data, routing, and MSME inclusion.	DPDP Act compliance; ULIP governance provisions; logistics data stewardship; interoperable trusted data-sharing mechanisms; MSME digital-access mechanisms; algorithmic accountability; sovereign oversight of operational logistics data (DPIIT, 2024; International Data Spaces Association [IDSA], 2024).	DPIIT–NCAER (2023); PM GatiShakti (2024)
S ³ Shock-Resilience (IV ₃)	Degree to which corridors maintain institutional continuity under disruption without governance regression.	Documented rerouting protocols; inter-ministerial crisis authority; cascade-isolation mechanisms.	Ministry of Railways (2023); expert interviews ($n = 19$)

Variable	Conceptual Definition	Governance Evidence Proxy	Source
Composite S-score	Composite governance band determined using a non-compensatory minimum-condition rule, whereby the weakest governance pillar defines overall institutional maturity because Sustainability (S^1), Sovereignty (S^2), and Shock-Resilience (S^3) are treated as jointly necessary governance conditions (Ragin, 2008; Schneider & Wagemann, 2012).	$(S^1+S^2+S^3)/3$; band = weakest pillar.	Derived
Transit Variability (DV ₁)	Unpredictability of freight transit times as an indicator of operational governance fragility.	World Bank LPI Timeliness indicator used as a contextual directional benchmark rather than a corridor-level performance measure; interpreted alongside corridor-specific governance evidence and expert validation (Arvis et al., 2023).	World Bank LPI (2023)
Recovery Duration (DV ₂)	Indicative duration to restore normal operations following disruption.	Absence of rerouting protocols; COVID-19 road-fallback documentation; expert assessment.	Expert interviews (n = 19)
Modal Fallback Intensity (DV ₃)	Degree of institutional exposure through reliance on carbon-intensive fallback modes under disruption.	Approximately 60% road dependency; 6–8% coastal modal share despite cost and carbon advantages.	DPIIT–NCAER (2023); PM GatiShakti National Master Plan (2021)

Source: Developed by the authors (2026), based on NITI Aayog (2023), DPIIT–NCAER (2023), Ministry of Railways (2023), PM GatiShakti (2024), World Bank LPI (2023), and expert interviews (n = 19).

Note: Governance evidence proxies are indicative institutional measures rather than direct operational performance indicators. Composite governance bands are derived using the S³-LOGIC non-compensatory minimum-condition rule, with contextual benchmarks supplemented by documentary evidence and expert validation.

4.6 Table 3 translates the conceptual architecture of S³-LOGIC into measurable governance variables by establishing clear links between theoretical constructs, observable evidence, and authoritative data sources. The operationalisation distinguishes governance capability from conventional logistics performance through

institution-based proxies capturing sustainability, sovereignty, and shock-resilience. Dependent variables emphasise operational vulnerability rather than efficiency outcomes, enabling assessment of institutional robustness under disruption. The inclusion of documentary evidence, contextual international benchmarks, and expert validation strengthens construct validity while ensuring methodological transparency and replicability. Overall, the table provides the empirical foundation for the comparative governance assessment and demonstrates how abstract governance concepts are systematically transformed into an auditable diagnostic framework. Table 4 defines the governance band classification criteria used to interpret S³-LOGIC scores by linking quantitative score ranges with corresponding institutional governance maturity.

Table 4. S³-LOGIC Governance Band Classification Criteria

Band	Score Range	Governance Evidence Pattern	Institutional Interpretation
Resilient	4.2–5.0	Governance provisions documented, operational, and institutionally tested across all sub-dimensions.	Institutional architecture credibly protects performance under disruption.
Consolidating	3.4–4.1	Most provisions present and improving; minor documentation or coordination gaps remain.	Governance maturity advancing but not yet fully institutionalised.
Transitional	2.6–3.3	Genuine positive trajectory on some sub-dimensions; critical provisions still absent or undocumented.	Progress is real but governance-constrained; gains are not yet institutionally protected.
Fragile	1.6–2.5	Governance provisions predominantly absent or undocumented across most sub-dimensions.	Institutional deficiency confirmed; governance failure probable during disruption without reform.
Critical	1.0–1.5	Active deterioration across multiple sub-dimensions; prior provisions being reversed or abandoned.	Severe institutional fragility; immediate intervention required.

Source: Developed by the authors (2026) based on the S³-LOGIC Governance Framework and configurational assessment principles of Ragin (2008) and Schneider and Wagemann (2012).

Note: Governance bands are theory-derived diagnostic classifications operating under the non-compensatory minimum-condition rule, whereby the weakest governance pillar determines the overall institutional maturity classification.

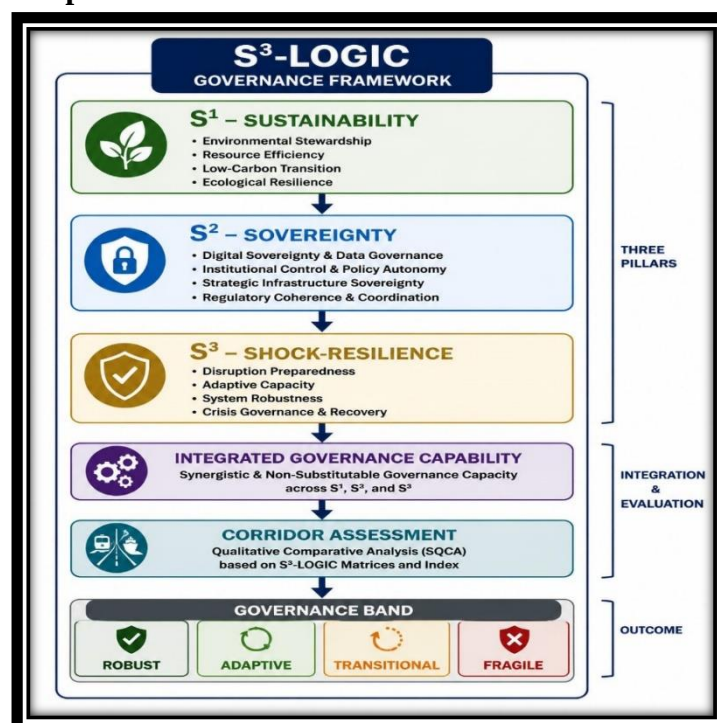
4.7 Table 4 operationalises governance maturity through five progressively ordered diagnostic bands, enabling consistent interpretation of corridor governance performance. By combining score thresholds with institutional evidence patterns and governance interpretations, the framework shifts assessment beyond numerical scoring to policy-relevant diagnosis. The non-compensatory classification logic ensures that critical institutional deficiencies remain visible, reinforcing the study's premise that governance effectiveness depends on balanced maturity across all three S³-LOGIC pillars rather than average performance.

5. The S³-LOGIC: Governance Diagnostic Architecture

5.1 S³-LOGIC is conceptualised as a governance diagnostic architecture: an institutional assessment instrument identifying where the governance conditions necessary for sustainable, sovereign, and resilient logistics outcomes are absent, underdeveloped, or institutionally visible. It differs from performance indices such as the Logistics Performance Index (LPI) in three respects. The framework is also conceptually aligned with the OECD Infrastructure Governance Indicators (OECD, 2023), extending their national-level governance principles to corridor-level institutional assessment by incorporating sustainability, digital sovereignty, and shock resilience as integrated governance conditions. First, it diagnoses governance architecture rather than logistics performance, assessing whether the institutional conditions necessary to sustain freight movement, preserve sovereign control over logistics data, and maintain long-term environmental sustainability are actually present. Second, it operationalises the non-substitutability principle by recognising that sustainability, sovereignty, and shock resilience constitute jointly the necessary governance capabilities whose absence cannot be compensated by strengths elsewhere and is consistent with the cumulative capability perspective of the Sand Cone Model and contemporary supply chain resilience research (Ferdows & De Meyer, 1990; Hohenstein et al., 2015). Third, it is explicitly designed for emerging-economy logistics systems undergoing simultaneous infrastructure expansion, digital transformation, and institutional development, where governance maturity rather than operational performance constitutes the binding constraint on long-run resilience.

5.1.1 Figure 2 presents the conceptual architecture of the S³-LOGIC Framework. It illustrates the three non-substitutable governance pillars—Sustainability (S¹), Sovereignty (S²), and Shock-Resilience (S³)—and their integration into a unified governance diagnostic for evaluating multimodal logistics systems operating under conditions of systemic disruption.

Figure 2. Conceptual Architecture of the S³-LOGIC Governance Framework



Source: Developed by the Authors (2026).

Note: The framework conceptualises Sustainability (S^1), Sovereignty (S^2), and Shock-Resilience (S^3) as jointly necessary and non-substitutable governance conditions. Their integrated assessment produces an overall governance maturity classification for multimodal logistics systems.

5.1.2 Figure 2 demonstrates that governance effectiveness emerges from the simultaneous interaction of the three institutional pillars rather than from excellence in any single dimension. The framework therefore evaluates governance capability as an integrated institutional construct, enabling the identification of vulnerabilities that conventional performance-based logistics indicators may not capture.

5.2 Pillar Definitions and Sub-Dimensions

5.2.1 **S¹—Sustainability Governance** assesses the degree to which governance operates within ecological constraints and actively manages demand-induced rebound risk across four sub-dimensions: **(a)** absolute environmental load trajectory-whether freight-volume growth is decoupling from emissions growth, or whether efficiency gains are being offset by demand-induced expansion; **(b)** energy-transition optionality-whether infrastructure investment embeds or forecloses low-carbon freight options; **(c)** demand-moderation governance-whether institutional mechanisms manage overall freight intensity rather than per-unit efficiency alone; and **(d)** modal sustainability profile-the governance-determined composition of the freight modal mix. These dimensions align with the strategic objectives of the National Logistics Policy (2022), PM GatiShakti, and Maritime India Vision 2030, operationalising sustainability as a measurable dimension of institutional governance rather than a purely environmental objective.

5.2.2 **S²—Sovereignty Governance** assesses the degree to which national institutions retain governance agency over the digital and organisational architecture of logistics operations while ensuring secure, interoperable, and accountable management of operational logistics data. Beyond sovereign control, the pillar evaluates whether governance arrangements provide trusted data-sharing mechanisms, algorithmic transparency, traceable data stewardship, interoperable data governance, and equitable digital participation for logistics stakeholders, particularly MSMEs, consistent with the governance principles of the International Data Spaces (IDS) architecture, which promotes sovereign, secure, and policy-controlled exchange of operational data across organisational boundaries (International Data Spaces Association [IDSA], 2024; Department for Promotion of Industry and Internal Trade [DPIIT], 2024). **(a)** Trusted data-governance architecture-whether operational logistics data are exchanged through secure, interoperable, and policy-controlled governance arrangements that preserve data sovereignty, traceability, and institutional accountability (IDSA, 2024) **(b)** governance agility-the institutional capacity to rapidly coordinate and reconfigure logistics operations across multiple agencies and jurisdictions under novel disruption conditions through established collaborative governance arrangements (Ansell & Gash, 2008) **(c)** crisis-command architecture-pre-authorised inter-ministerial coordination authority supported by formally designated lead institutions and documented cross-agency coordination protocols for rapid operational decision-making during logistics disruptions, consistent with collaborative governance and inter-

organisational coordination theory (Ansell & Gash, 2008) **(d)** MSME inclusion governance-whether micro, small and medium operators retain meaningful, protected participation in digitalised freight networks.

5.2.3 **S³—Shock-Resilience Governance** assesses the degree to which corridors maintain institutional continuity under disruption without governance regression across four sub-dimensions that operationalise an agile–adaptable–aligned resilience vocabulary at corridor level, consistent with OECD systemic resilience principles ([OECD 2024](#)) **(a)** disruption-absorption capacity; **(b)** governance agility-the speed and authority with which institutions can reconfigure operations under novel disruption profiles; **(c)** crisis-command architecture-pre-authorised inter-ministerial coordination authority; and **(d)** documented rerouting and cascade-isolation protocols. The practical relevance of these sub-dimensions was demonstrated during the 2024–2025 Red Sea crisis or recent Suez Canal crisis which required coordinated decision-making across shipping, customs, ports, railways, and logistics authorities. Such institutional continuity depends upon pre-defined cross-agency governance arrangements, designated lead authorities, and formal coordination mechanisms rather than ad hoc administrative responses, consistent with collaborative governance theory (Ansell & Gash, 2008).

5.2.4 Consistent with SQCA's configurational logic, these twelve sub-dimensions are assessed as interacting institutional conditions rather than independent additive indicators: different combinations of strengths and deficiencies may produce comparable vulnerability profiles, while material weakness in any critical sub-dimension may constrain overall resilience depending on the broader institutional configuration. Score-band thresholds were established deductively from the framework's theoretical structure and subsequently refined through expert validation. Consistent with the logic of necessary conditions in Qualitative Comparative Analysis, the composite governance band is determined by the lowest-performing pillar because the absence of any one governance capability constrains overall institutional maturity irrespective of strengths elsewhere (Ragin, 2008; Schneider & Wagemann, 2012). Consequently, the thresholds function as theory-informed diagnostic classifications rather than statistically calibrated intervals.

6. Empirical Application: Three Corridor Governance Profiles

6.1 Western Dedicated Freight Corridor (WDFC): Industrial Logistics Governance

6.1.1 The WDFC spans 1,504 km from Dadri to JNPT, serving India's primary industrial and manufacturing freight flows. The WDFC, therefore, provides the most demanding test of this paper's central argument that infrastructure investment, however substantial, does not produce governance architecture as a by-product.

6.1.2 **S¹—Transitional (3.2).** The sustainability trajectory is genuinely positive as modal shift from road to rail is advancing and 25 kV AC electrification eliminates diesel dependency on the rail segment with the commercial cost differential creating structural incentives for continued shift. Yet a critical governance provision is absent. The National Logistics Policy (2022) promotes efficiency and multimodal transport but establishes no explicit mechanism for monitoring absolute environmental load or demand-induced rebound. The Transitional designation therefore reflects genuine progress constrained by an identifiable governance gap, not a superficial assessment.

6.1.3 **S²—Fragile (1.8).** Although the Unified Logistics Interface Platform (ULIP) and the Logistics Data Bank have substantially advanced digital integration, no publicly documented governance framework currently specifies operational responsibilities for data stewardship, policy-controlled data sharing, algorithmic accountability, or sovereign oversight of routing and origin–destination logistics data, as advocated in emerging International Data Spaces (IDS) governance models (Department for Promotion of Industry and Internal Trade [DPIIT], 2024; International Data Spaces Association [IDSA], 2024). The predominance of small and fragmented road freight operators in first- and last-mile connectivity continues to constrain digital integration with corridor booking systems, limiting institutional visibility over operational logistics data and reducing MSME participation in formal digital logistics networks (NCAER, 2023; Ministry of Road Transport and Highways, 2023). As platform adoption deepens without sovereign governance provisions, commercial operators are gaining institutional control over critical logistics data—a condition invisible in freight speed or cost metrics.

6.1.4 **S³—Fragile (1.9).** No publicly documented inter-ministerial disruption-response protocol exists for the WDFC. Crisis-governance authority is distributed across DFCCIL, the Ministry of Railways, DPIIT, and state transport departments without a joint coordination architecture. [Boin et al., \(2017\)](#) find that ad hoc coordination under disruption is systematically less effective than pre-established institutional protocols, a finding directly illustrated by the COVID-19 experience and WDFC rail disruptions triggered road-freight reversion within days, imposing carbon-intensity penalties, rate spikes, and MSME cost burdens that documented governance protocols could have mitigated.

6.1.5 **Composite—Fragile (2.30).** The minimum-condition rule assigns a Fragile composite, governed by the S² and S³ scores. The critical analytical point is what the 0.40-point gap between S¹ (Transitional) and the weakest pillars (Fragile) reveals: India's most capital-intensive freight corridor has advanced its operational sustainability profile while leaving the institutional architecture protecting that profile against platform dependency and disruption largely

unconstructed. This exposure has become more consequential as the WDFC increasingly supports internationally integrated supply chains under initiatives such as the India–Middle East–Europe Economic Corridor (IMEC).

6.2 Eastern Dedicated Freight Corridor (EDFC): Agricultural and Welfare Logistics Governance

6.2.1 At 1,873 km from Ludhiana to Dankuni the EDFC serves agricultural and bulk commodity flows, including PDS grain movements supporting an estimated 813.5 million beneficiaries under the National Food Security Act and the Pradhan Mantri Garib Kalyan Anna Yojana. Its governance context differs qualitatively from the WDFC's as the EDFC sits at the intersection of commercial logistics, state food-procurement mandates, and national food security obligations. Governance failure here carries welfare consequences that extend well beyond commercial disruption, making the EDFC profile analytically distinct rather than simply a lower score on the same scale.

6.2.2 **S¹—Transitional (3.1)**. The trajectory is marginally weaker than the WDFC's, reflecting the distinct constraints of agricultural freight: Public Distribution System (PDS) origin-destination pairs are fixed by cultivation geography and procurement mandates rather than commercial optimisation, reducing the institutional degrees of freedom available for sustainability intervention, while cold-chain governance for temperature-sensitive commodities remains underdeveloped.

6.2.3 **S²—Fragile (1.7)**, the weakest S² score in the study. Sovereignty governance here is institutionally more complex than the WDFC's as PDS coordination involves the Food Corporation of India, state civil-supplies corporations, private transport contractors, and multiple state agencies, each with distinct jurisdictional interests and data arrangements. While an overarching framework for logistics sector data sovereignty remains absent the MSME-dominated agricultural logistics operators lack structured digital access to formal booking platforms.

6.2.4 **S³—Fragile (1.8)**, No documented rerouting protocols exist for PDS grain flows under disruption. COVID-19 forced PDS movements onto road transport at spot rates, generating supply delays and cost spikes affecting national food distribution reliability, and the 2024–2025 Red Sea crisis similarly demonstrated that external geopolitical disruptions can affect food-security logistics through prolonged transit times and shipping uncertainty. Unlike commercial freight disruption, where the primary consequence is financial, EDFC resilience governance failure has direct public welfare implications.

6.2.5 **Composite—Fragile (2.2)**. Institutional complexity here becomes a resilience risk when governance responsibility remains fragmented without corresponding coordination architecture, consistent with the collaborative governance framework of [\(Ansell & Gash, 2008\)](#). The multi-stakeholder architecture is institutionally necessary but distributes authority without creating the coordination structure required for collective resilience governance.

6.3 Coastal Maritime Corridor: Governance-Suppressed Sustainability

6.3.1 India's 7,517-km coastline connects major commercial ports through the Sagarmala development framework, and coastal maritime freight carries the lowest carbon intensity of any available long-distance mode. Despite sustained investment under the Sagarmala Programme, coastal shipping continues to account for a relatively modest share of India's domestic freight movement. Recent policy assessments attribute this less to physical infrastructure constraints than to persistent governance challenges involving multimodal integration, regulatory coordination, and institutional activation of coastal logistics (Ministry of Ports, Shipping and Waterways, 2024; NITI Aayog, 2024). The analytical challenge presented by this corridor is distinctive: although several coastal states have achieved substantial progress in port modernisation and logistics integration, nationally consistent governance arrangements for multimodal coordination, environmental stewardship, and digital logistics integration remain uneven, constraining the corridor's full institutional potential.

6.3.2 S¹—Transitional (3.0). The Sagarmala Programme, Maritime India Vision 2030, and Amrit Kaal Vision 2047 have directed substantial investment toward port modernisation and multimodal integration, but these initiatives remain principally infrastructure- and capacity-oriented. Institutional mechanisms governing coastal modal activation, environmental performance monitoring, and workforce capability development remain comparatively underdeveloped and so the sustainability performance gap is essentially a governance gap the physical and commercial preconditions are already present.

6.3.3 S²—Fragile (1.6), the lowest sovereignty score in the study. Port concession agreements at major terminals have expanded private operational responsibilities without publicly documented governance provisions for policy-controlled operational data sharing, traceability, institutional data stewardship, or sovereign oversight, despite emerging International Data Spaces (IDS) governance principles advocating interoperable and trusted data-sharing architectures for critical infrastructure (International Data Spaces Association [IDSA], 2024; Department for Promotion of Industry and Internal Trade [DPIIT], 2024).

6.3.4 S³—Fragile (1.7). Port closures triggered by cyclonic events, maritime security incidents, or industrial action produce immediate road-freight reversion with no documented inter-port rerouting protocol. India's multi-port coastal architecture theoretically offers natural resilience optionality between ports when one is disrupted, but the absence of inter-port rerouting governance, standardised container equipment pools, and rapid modal-switch coordination means this architectural optionality cannot be operationally exercised.

6.3.5 Composite—Fragile (2.10), the lowest composite score in the study, with the weakest individual scores in both S² and S³. That India's most environmentally and economically advantageous freight mode is also its least institutionally supported and is analytically consequential as it reveals that governance omission, not infrastructure deficit, is the primary constraint on India's logistics sustainability performance. Critically, this constraint is governance-reversible rather than requiring the capital-investment timelines of physical infrastructure development.

7. Results

7.1 This section presents the empirical application of the S³-LOGIC framework by comparing governance maturity across the three representative freight corridors and examining their directional association with operational vulnerability. Table 5 summarises the comparative S³-LOGIC governance assessment, presenting pillar scores, composite governance bands, and their directional relationship with corridor vulnerability.

Table 5. Master S³ Governance Diagnostic and Directional Vulnerability Association

Corridor	S ₁	S ₂	S ₃	Composite	Band	Composite Vulnerability*	Association
Western DFC (Industrial/Manufacturing)	3.2	1.8	1.9	2.30	Fragile	3.80	↓ Inverse
Eastern DFC (Agricultural/Bulk)	3.1	1.7	1.8	2.20	Fragile	4.03	↓ Inverse
Coastal Maritime (Export/Coastal)	3.0	1.6	1.7	2.10	Fragile	4.30	↓ Inverse

Source: Developed by the authors (2026) from S³-LOGIC diagnostic assessment, documentary evidence, and expert validation (n = 19)

Note: Composite governance bands are derived using the non-compensatory minimum-condition rule, while composite vulnerability represents the average of Transit Variability, Recovery Duration, and Modal Fallback Intensity on a 1–5 scale

7.2 Table 5 reveals that all three corridors remain institutionally Fragile despite differing sustainability scores. Lower governance maturity consistently corresponds with higher operational vulnerability, reinforcing the study's proposition that sovereignty and shock-resilience constitute the principal constraints on resilient multimodal logistics governance.

7.3 Although the Western Dedicated Freight Corridor records the strongest sustainability performance ($S^1 = 3.2$), its composite governance band remains *Fragile* because weaknesses in Sovereignty ($S^2 = 1.8$) and Shock-Resilience ($S^3 = 1.9$) constitute binding institutional constraints under the framework's minimum-condition logic. This illustrates the framework's central proposition that governance maturity depends on the integrity of all three pillars rather than on average performance. Table 6 positions India's logistics governance within an international context by comparing selected structural, operational, and governance indicators with China and the European Union.

Table 6. India Logistics Governance — Comparative International Context

Indicator	India (2023)	China (2023)	EU Benchmark (2023)
Logistics cost as % of GDP	~8–9% (NCAER, 2024)	~14–15%	~8–10%
Rail freight modal share	~27%	~47%	Germany ~28%; EU average ~18%

Indicator	India (2023)	China (2023)	EU Benchmark (2023)
Coastal/waterway freight share	<10% of domestic freight; despite substantial infrastructure expansion under the Sagarmala Programme, multimodal integration and governance coordination continue to constrain modal shift (Ministry of Ports, Shipping and Waterways, 2024).	~47% (including inland waterways)	~13% (EU short-sea average)
World Bank LPI rank (2023)	38th / 139	19th / 139	Netherlands 1st; Germany 3rd
Logistics data governance and sovereignty	DPDP Act provides a general legal framework, while the Unified Logistics Interface Platform (ULIP) has substantially advanced digital integration. However, publicly documented sector-specific governance provisions for operational logistics data stewardship, interoperable data sharing, algorithmic accountability, and sovereign oversight remain limited (DPIIT, 2024).	Operational logistics-data governance supported by the Data Security Law and sector-specific regulatory oversight.	GDPR together with NIS2 and emerging common European data-space governance frameworks support trusted operational data sharing.
Dedicated freight crisis-coordination authority	No dedicated statutory freight crisis authority	Present (Ministry of Transport coordination)	Varied; national frameworks operational

Source: Developed by the authors (2026), based on NCAER (2024), World Bank LPI (2023), Ministry of Ports, Shipping and Waterways (2024), DPIIT (2024), China's Data Security Law, and EU GDPR and NIS2 frameworks

Note: The comparison is indicative rather than evaluative and is intended to contextualise India's governance maturity against representative international logistics systems using publicly available policy and performance indicators.

7.4 Table 6 indicates that India has achieved competitive logistics costs but continues to lag behind China and established European systems in modal diversification, digital logistics governance, and institutional crisis coordination. The comparison reinforces the study's central argument that future logistics competitiveness depends not only on infrastructure expansion but also on strengthening governance capacity, sovereign data frameworks, and adaptive institutional resilience

8. Discussion

8.1 The finding that every corridor, including India's most capital-intensive freight investment, receives a Fragile composite governance assessment reveals a structural pattern that institutional economics has long theorised but logistics policy has been slow to operationalise: physical infrastructure and governance architecture are complementary institutional requirements, not substitutes. Infrastructure creates physical capacity for performance improvement; governance architecture determines whether that capacity is institutionally controlled, ecologically sustainable, and operationally resilient under disruption. Multi-level governance theory (Hooghe & Marks, 2003) explains why this asymmetry is a structural property of the intergovernmental architecture rather than simply a policy design failure. India's freight corridor governance exhibits pronounced 'Type II' characteristics like polycentric, task-specific, overlapping jurisdictions without a single overarching authority all under which governance gaps persist not because actors lack awareness or intent, but because no single institutional actor possesses the authority and information required to govern effectively across all dimensions alone. This pattern is consistent with contemporary theories of cross-jurisdictional infrastructure governance, where institutional fragmentation persists despite shared policy objectives unless supported by formal coordination mechanisms and designated lead authorities (Hooghe & Marks, 2003; Ansell & Gash, 2008).

8.2 The S³-LOGIC prescriptions, particularly the proposed National Freight Corridor Resilience Authority (G3) and National Logistics Data Sovereignty Act (G1), are accordingly framed as institutional interventions designed to create 'Type I' authority where 'Type II' fragmentation currently prevents coordinated governance action. These governance prescriptions complement internationally recognised infrastructure-governance principles by strengthening institutional coordination, strategic oversight, resilience, and accountability in ways consistent with the OECD Recommendation on the Governance of Infrastructure and the OECD Infrastructure Governance Indicators, while extending their application to corridor-level logistics governance in emerging economies (OECD, 2020; OECD, 2023). Notably, the composite scores of the WDFC and the Coastal Maritime Corridor (India's highest-investment and least institutionally supported freight systems, respectively) differ by only 0.20 points, illustrating that, although several Indian states have achieved substantial advances in logistics implementation and infrastructure delivery, sustained infrastructure investment without commensurate institutional governance development continues to constrain the durability and scalability of these gains at the national multimodal logistics level (NITI Aayog, 2024).

8.3 While all three corridors receive Fragile composite designations, their vulnerability profiles are institutionally differentiated in ways with direct implications for governance priority-setting. The WDFC's exposure is primarily commercial and digital: as DFC digitalisation deepens, the principal risk is a progressive, increasingly hard-to-reverse transfer of governance influence over logistics data to commercial platform operators. The EDFC's exposure is institutionally and socially more complex, since its multi-stakeholder architecture compounds data sovereignty and resilience gaps, and its resilience fragility carries a welfare dimension (public food-security consequences, not merely commercial disruption) that should elevate its institutional priority. The Coastal Maritime Corridor's exposure is structurally distinct from both DFCs, reflecting a governance-induced mode-suppression dynamic whereby multimodal governance integration has not progressed at the same pace as infrastructure investment under the Sagarmala Programme, limiting the operational uptake of coastal

shipping despite its inherent sustainability advantages (Ministry of Ports, Shipping and Waterways, 2024).

8.4 These fragilities carry strategic implications beyond operational logistics management. The post-2020 geopolitical environment or the present day Suez Canal crisis has rendered logistics infrastructure a domain of active strategic competition, and the S² Sovereignty dimension is directly implicated in how that competition affects national strategic autonomy. Where critical freight-routing decisions are governed by platform algorithms rather than national institutional frameworks, the possibility that platform governance could be influenced by geopolitical pressure on operators' home jurisdictions represents a non-trivial institutional risk and not a speculative one. Since COVID-19 already demonstrated that platform capacity-allocation decisions under shortage conditions had distributional consequences that national governments were institutionally unequipped to challenge. The maritime dimension is particularly acute given Red Sea and Suez Canal security incidents, Indian Ocean power competition, and Strait of Malacca vulnerability, none of which has a coordinated governance response at the national freight-system level. India's geographically distributed port system provides substantial latent resilience; however, this resilience cannot be fully operationalised without formal multi-jurisdictional coordination mechanisms, designated lead institutional authority, and pre-defined inter-port rerouting governance capable of synchronising decision-making across ports, customs, railways, and shipping agencies during disruption (Ansell & Gash, 2008). Beyond physical coordination, digital coordination has become equally important as logistics platforms increasingly rely on algorithmic routing and automated operational decisions. A related and rapidly emerging dimension concerns artificial intelligence: as AI systems take on greater roles in freight matching, route optimisation, and predictive logistics management, institutional provisions governing algorithmic transparency, accountability, and sovereign oversight remain comparatively underdeveloped despite India's broader national AI policy foundations a sequencing priority with strategic implications equivalent to those of data sovereignty.

8.5 The non-substitutability principle receives consistent directional support from the empirical application. In every corridor, the Transitional S¹ trajectory is institutionally constrained by Fragile S²/S³ designations, producing a Fragile composite despite genuine sustainability progress, in line with the theoretical proposition that governance robustness requires conjunctive institutional conditions. A one-at-a-time sensitivity analysis was undertaken by varying each individual pillar score independently by ± 0.2 while holding the remaining pillar scores constant. Under all simulated combinations, the composite governance band remained unchanged because the minimum-condition rule assigns the overall governance classification according to the lowest-performing pillar. This non-compensatory approach is consistent with the logic of necessary conditions in Qualitative Comparative Analysis, whereby the absence of a single indispensable governance capability constrains overall institutional maturity irrespective of strengths elsewhere (Ragin, 2008; Schneider & Wagemann, 2012).

8.6 The policy significance of these findings extends beyond the three corridor archetypes examined in this study. As logistics systems become increasingly digital, interconnected, and geopolitically exposed, governance itself emerges as a strategic determinant of national competitiveness and economic security. Investments in infrastructure alone cannot guarantee resilient logistics outcomes if they are not supported by corresponding investments in institutional capability, regulatory coordination, data governance, and crisis-response architecture. Consequently, future

logistics policy should be evaluated not only by the physical assets it creates, but also by the governance capacity it institutionalises. This perspective reframes logistics governance from a predominantly operational concern into a strategic public-policy function that integrates economic efficiency, national sovereignty, environmental sustainability, and systemic resilience within a single governance architecture. The seven governance prescriptions proposed in this study should therefore be interpreted as mutually reinforcing institutional reforms rather than independent policy interventions, collectively strengthening the long-term adaptive capacity of India's multimodal logistics ecosystem.

9. Governance Prescriptions

9.1 The corridor diagnostics yield seven governance prescriptions, ordered by institutional urgency and structured to strengthen cross-jurisdictional governance capacity rather than infrastructure provision alone. Collectively, the prescriptions establish designated institutional responsibilities, interoperable governance arrangements, and coordinated decision-making mechanisms required to operationalise resilient multimodal freight governance across multiple agencies and jurisdictions (Ansell & Gash, 2008; OECD, 2023). Each targets a specific governance deficit identified through SQCA, proposes a concrete institutional intervention, and identifies the mechanism required for implementation (Table 7). The prescriptions are explicitly framed as complements to, not replacements for, existing legislation and strategy. While the Digital Personal Data Protection Act, 2023 establishes a general legal framework for personal data protection, it does not define sector-specific institutional responsibilities for operational logistics data stewardship, trusted inter-organisational data sharing, algorithmic accountability, or sovereign oversight of strategically significant freight information. The proposed National Logistics Data Sovereignty Act is therefore intended to complement, rather than replace, the DPDP Act by addressing governance functions unique to the logistics sector (DPIIT 2024; IDSA 2024), including the Digital Personal Data Protection Act, 2023, PM GatiShakti, Maritime India Vision 2030, and Amrit Kaal Vision 2047. Table 7 gives the seven prescriptions which are sequenced according to institutional dependency. Governance-enabling measures (G1–G3) establish the legal, organisational, and coordination foundations upon which operational resilience measures (G4–G6) and long-term sustainability interventions (G7) can be effectively implemented, reflecting the cumulative capability logic underpinning the S³-LOGIC framework (Ferdows & De Meyer, 1990). The convergence objective across all seven is a composite governance-band upgrade from Fragile to Transitional by 2028–2030, consistent with the UN Decade of Sustainable Transport's implementation window.

Table 7. S³-LOGIC Governance Reform Programme - Prioritised Institutional Prescriptions

No.	Pillar	Governance Gap	Intervention	Horizon
G1	S ²	Absent national logistics data sovereignty framework; routing, pricing, and cargo data ceded to platform operators by institutional default.	Enact a National Logistics Data Sovereignty Act as a sector-specific complement to the Digital Personal Data Protection Act, 2023, establishing statutory governance provisions for operational logistics data stewardship, interoperable trusted data sharing, algorithmic accountability, routing-data governance, and sovereign oversight of strategically significant freight information within ULIP platforms. The proposed legislation complements the DPDP Act by addressing governance functions unique to multimodal logistics that are not explicitly covered under the general personal data protection framework (Department for Promotion of Industry and Internal Trade [DPIIT], 2024;).	12 months
G2	S ²	Structural exclusion of MSME operators from formal digital logistics networks.	Mandate MSME-accessible APIs integrated with the ULIP, enabling interoperable access to publicly funded logistics platforms through simplified onboarding, vernacular interfaces, standardised API protocols, and credit-linked digital access to reduce barriers to formal digital logistics participation (DPIIT, 2024).	18 months
G3	S ³	No inter-ministerial crisis coordination authority; disruption governance distributed without joint response mandate.	Establish a National Freight Corridor Resilience Authority (NFCRA), aligned with the OECD Recommendation on the Governance of Infrastructure (2020) and informed by the OECD Infrastructure Governance Indicators (2023), to provide statutory cross-agency coordination, designated lead authority, and pre-defined multimodal rerouting protocols for nationally significant freight disruptions.	18 months

No.	Pillar	Governance Gap	Intervention	Horizon
G4	S ¹ /S ³	Lowest unit cost and carbon intensity mode remains institutionally suppressed.	Coastal Freight Governance Programme integrated with the Sagarmala Programme and Maritime India Vision 2030, strengthening multimodal coordination, cabotage optimisation, digital port coordination, and coastal logistics governance to improve the operational utilisation of existing maritime infrastructure (Ministry of Ports, Shipping and Waterways, 2024; Maritime India Vision 2030).	Phase 1: 2026–2028
G5	S ¹	Demand moderation absent from NLP 2022; efficiency gains exposed to demand-induced rebound.	Introduce a Freight Demand Moderation Instrument through amendments to the National Logistics Policy (2022), establishing absolute environmental-load monitoring and freight-intensity benchmarks to complement efficiency indicators. Similar demand-management approaches are increasingly employed in transport decarbonisation and sustainable freight policy to manage rebound effects while preserving logistics efficiency gains (International Transport Forum, 2023; OECD, 2023).	Policy cycle 2026–2030
G6	S ²	Platform concentration advancing without governance oversight.	Annual logistics platform governance audits assessing market concentration, data governance compliance, and MSME digital participation.	24 months
G7	S ² /S ³	Port data sovereignty absent at coastal terminals; concession agreements cede operational data without governance provisions.	Operational logistics data governance clauses incorporated into all new port concession agreements, consistent with the DPDP Act, 2023.	12–18 months

Source: Developed by the authors (2026), based on S³-LOGIC diagnostic findings and synthesised from DPIIT (2024), OECD (2020; 2023), Ministry of Ports, Shipping and Waterways (2024), Maritime India Vision 2030, and the Digital Personal Data Protection Act, 2023.

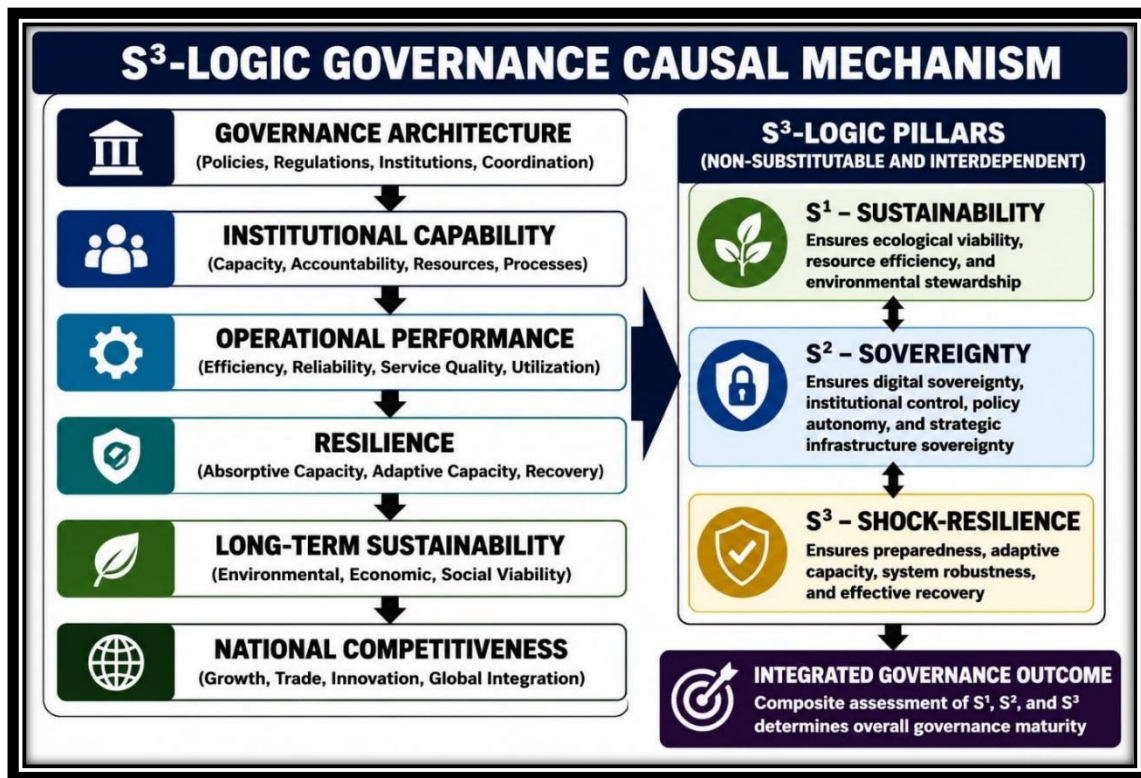
Note: The prescriptions are prioritised according to the governance vulnerabilities identified through the S³-LOGIC assessment and are designed to complement existing national logistics policies through phased institutional implementation rather than replace them.

9.2 Table 7 translates the S³-LOGIC diagnostic into a sequenced governance reform agenda by aligning identified institutional deficiencies with targeted policy interventions and realistic implementation horizons. The recommendations prioritise strengthening digital sovereignty, crisis-governance architecture, multimodal coordination, sustainable freight governance, and platform accountability before

addressing longer-term structural reforms. The phased implementation strategy recognises differences in institutional complexity and legislative requirements while ensuring that reforms remain operationally feasible. Collectively, the prescriptions reinforce the framework's practical orientation by demonstrating how governance vulnerabilities identified through the three pillars can be systematically converted into actionable policy measures, thereby bridging the gap between governance diagnosis and institutional reform.

9.3 Figure 3 illustrates the causal governance mechanism underpinning the S³-LOGIC Framework. It explains how institutional governance capability influences logistics performance, resilience, and long-term competitiveness through the integrated operation of the three governance pillars.

Figure 3. Causal Governance Mechanism of the S³-LOGIC Framework



Source: Developed by the Authors (2026).

Note: The framework conceptualises Sustainability (S¹), Sovereignty (S²), and Shock-Resilience (S³) as interdependent governance conditions that collectively strengthen institutional capability and improve system-wide logistics outcomes under conditions of uncertainty.

9.4 Figure 3 illustrates the theoretical logic of S³-LOGIC by demonstrating that governance quality operates as the primary causal mechanism linking institutional capability to operational performance, resilience, sustainability, and long-term competitiveness. The framework emphasises that governance effectiveness emerges from the integrated interaction of all three pillars rather than from improvements in any single governance dimension.

9.5 Taken together, the reform programme demonstrates that infrastructure modernisation alone is insufficient for resilient multimodal logistics governance unless accompanied by institutional sovereignty, sector-specific data governance that complements the Digital Personal Data Protection Act, 2023, and adaptive disruption-management mechanisms. The comparative assessment therefore validates the study's central proposition: governance asymmetries, not infrastructure deficits, continue to constrain India's logistics transformation trajectory, a conclusion reinforced by the 2024–2025 Red Sea crisis, during which prolonged shipping delays and route diversions demonstrated that corridor resilience increasingly depends on institutional coordination and adaptive governance rather than infrastructure capacity alone.

10. Theoretical Contributions: This paper advances six contributions to the governance and logistics infrastructure literature.

10.1 First, S³-LOGIC is advanced as a governance diagnostic architecture that integrates ecological viability, institutional sovereignty, and adaptive resilience as co-equal, non-substitutable governance conditions within a single instrument for emerging-economy multimodal infrastructure systems, connecting governance theory (Kettl, 2002; Ansell & Gash, 2008), resilience governance (Boin et al., 2017; Folke et al., 2010), digital sovereignty scholarship ([Floridi, 2020](#)), and infrastructure resilience research into an applied institutional diagnostic.

10.2 Second, the non-substitutability principle is formalised as an operational assessment rule, grounded in the cumulative capability perspective of the Sand Cone Model and contemporary supply chain resilience theory. Consistent with these perspectives, Sustainability (S¹), Sovereignty (S²), and Shock-Resilience (S³) are conceptualised as jointly necessary institutional capabilities whose absence cannot be compensated by strengths elsewhere, thereby providing the theoretical basis for the framework's minimum-condition rule (Ferdows & De Meyer, 1990; Hohenstein et al., 2015; Ragin, 2008). Existing composite indices, including the LPI, aggregate dimensions using weighted averages that allow high scores in some dimensions to mask critical deficits in others the minimum-condition rule prevents this masking, and the empirical application demonstrates the principle operating in practice, since Transitional S¹ scores are governance-constrained by Fragile S²/S³ designations in every corridor.

10.3 Third, a governance-first paradigm is advanced for emerging-economy logistics, arguing that institutional governance capacity should develop in parallel with infrastructure investment to sustain long-term system resilience. This perspective is consistent with multi-level and collaborative governance theories, which emphasise that infrastructure performance depends upon effective coordination across institutions and jurisdictions rather than physical investment alone (Hooghe & Marks, 2003; Ansell & Gash, 2008). The empirical finding that India's highest-investment and least institutionally supported corridors sit only 0.20 composite points apart supports this paradigm directly.

10.4 Fourth, the mode-suppression dynamic is identified as a generalisable diagnostic construct describing situations in which governance integration, multimodal coordination, and institutional activation lag behind infrastructure investment, thereby constraining the utilisation of an operationally viable and environmentally preferable freight mode. Recent implementation experience under the Sagarmala Programme indicates that governance coordination has become a more significant constraint than physical infrastructure availability for expanding coastal freight utilisation (Ministry of Ports, Shipping and Waterways, 2024). Applicable to any logistics system in which access to a superior mode is structurally constrained by institutional barriers.

10.5 Fifth, S³-LOGIC is positioned as a technology-neutral, institutionally specific governance diagnostic applicable at both national and sub-national levels, recognising that logistics governance maturity varies across jurisdictions. This flexibility supports implementation of the UN Decade of Sustainable Transport while accommodating institutional variation identified across Indian states (NITI Aayog, 2024), which requires governance diagnostics capable of translating aspiration into implementation at the national logistics planning level.

10.6 Sixth, S³-LOGIC advances a behavioural governance hypothesis grounded in measurement theory: that visible, institutionally maintained diagnostics mobilise governance investment in dimensions currently unmeasured and therefore under-prioritised. Merry's (2011) account of indicator governance documents the mechanism that indicators do not merely describe governance conditions but actively constitute them by making previously invisible institutional gaps legible to policymakers in forms that trigger accountability and resource-allocation responses, a mechanism for which the LSCI's documented port-investment effect provides an empirical precedent in a directly analogous logistics context.

10.7 Table 8 synthesises the principal theoretical contributions of the S³-LOGIC framework by linking identified diagnostic gaps with corresponding conceptual innovations, empirical evidence, and governance prescriptions.

Table 8. S³-LOGIC Theoretical Contribution Matrix

Existing Theory	Diagnostic Gap Addressed	S ³ -LOGIC Contribution	Evidence in This Paper	Prescription Link
Institutional isomorphism — DiMaggio & Powell (1983)	No instrument translates isomorphism theory into a measurable, corridor-level governance assessment.	Makes the governance architecture lag produced by isomorphic pressures visible and scored.	All three corridors receive Fragile S ² /S ³ despite strong infrastructure investment.	G1, G3
Resilience governance — Boin et al. (2016); Folke et al., 2010	No scoreable instrument applies resilience governance theory at sub-national corridor resolution.	Operationalises resilience governance as four scored sub-dimensions applied consistently across corridors.	S ³ = 1.7–1.9 across all corridors; independently corroborated by expert validation (n = 19).	G3
Digital sovereignty — Floridi (2020); Zuboff (2019)	No diagnostic instrument assesses institutional control over routing algorithms, cargo data, and platform concentration.	Operationalises digital sovereignty as four scored S ² sub-dimensions at the corridor-institutional level.	S ² = 1.6–1.8, the lowest pillar across all corridors.	G1, G6, G7

Existing Theory	Diagnostic Gap Addressed	S ³ -LOGIC Contribution	Evidence in This Paper	Prescription Link
Dynamic capabilities Beske (2012)	No instrument applies sensing/seizing/reconfiguring capability theory to public logistics governance at corridor scale.	S ³ sub-dimensions map directly onto sensing (disruption absorption), seizing (crisis command), and reconfiguring (governance agility).	Absence of all four S ³ sub-dimensions confirmed across all three corridors by SQCA and expert validation.	G3
Collaborative governance — Ansell & Gash (2008); Emerson et al. (2011)	No instrument assesses whether collaborative governance architecture for freight disruption response actually exists.	Crisis-command and governance-agility sub-dimensions assess pre-construction of collaborative response architecture.	COVID-19 documentation shows ad hoc coordination failure across all three corridors.	G3

Source: Developed by the authors (2026), synthesised from DiMaggio and Powell (1983), Folke et al. (2010), Boin et al. (2016), Floridi (2020), Zuboff (2019), Beske (2012), Ansell and Gash (2008), and Emerson et al. (2011)

Note: The matrix is interpretive and demonstrates how S³-LOGIC extends established governance theories by operationalising previously unmeasured institutional constructs into an empirically validated corridor-level governance diagnostic framework.

10.8 Table 8 demonstrates that S³-LOGIC contributes primarily through theoretical integration and operationalisation rather than introducing entirely new governance concepts. Existing governance theories are systematically translated into measurable diagnostic constructs capable of identifying institutional deficiencies at the corridor level. Empirical evidence from the three corridor assessments validates these theoretical extensions by linking observed governance gaps with specific policy prescriptions. The matrix further illustrates that institutional isomorphism, resilience governance, digital sovereignty, dynamic capabilities, and collaborative governance collectively underpin the framework's explanatory power. Consequently, S³-LOGIC bridges the long-standing gap between abstract governance theory and practical institutional diagnosis, strengthening both its conceptual coherence and applied policy relevance.

11. Limitations

11.1 Six limitations establish the appropriate interpretive scope for these findings. First, the assessment represents a temporal governance snapshot; as India's governance architecture continues to evolve through legislative reform and in response to disruptions such as the 2024–2025 Red Sea crisis, Suez Canal disruptions and governance maturity should be read as time-contingent rather than institutionally fixed. Second, the small-n case structure (three corridor archetypes) provides sufficient variation for theory-building diagnostic comparison but precludes inferential statistical generalisation; the appropriate evidential claim is theoretical transferability, not statistical representativeness. Third, despite predefined decision rules, explicit calibration criteria, and expert validation ($n = 19$), SQCA remains an interpretive configurational methodology in which calibration thresholds and rule choices are theory-informed rather than statistically determined. (Ragin, 2008; Ragin, 2014; Schneider & Wagemann, 2012). Fourth, S³-LOGIC is an emerging framework applied here for the first time across multiple corridor archetypes and has not yet undergone the iterative validation cycles that establish methodological maturity in established composite indices; findings should be read as diagnostic assessments with governance validity rather than definitive institutional measurements. Fifth, dependent-variable indicators are derived primarily from nationally aggregated secondary data rather than direct corridor-level governance measurement. Consequently, the framework does not capture sub-national variation in institutional maturity, regulatory implementation, or logistics governance performance across individual states and corridors. Sixth, governance scoring retains an irreducible interpretive element: independent application of the framework may yield slightly varied interpretations of the evidence.

11.2 Boundary Conditions

11.2.1 S³-LOGIC is designed for multimodal freight-corridor systems in emerging economies undergoing rapid, simultaneous infrastructure investment, logistics digitalisation, and exposure to compound disruption, and its applicability is bounded accordingly. While existing assessment frameworks—including the World Bank Logistics Performance Index (LPI), the UNCTAD Port Performance Scorecard, and the OECD Infrastructure Governance Indicators—provide valuable evidence on logistics performance, port operations, and national governance capacity, they do not explicitly diagnose the institutional interaction among sustainability, digital sovereignty, and shock resilience at the multimodal freight-corridor level. S³-LOGIC therefore complements, rather than replaces, these frameworks by providing a governance-diagnostic perspective specifically designed for institutional maturity assessment in emerging-economy logistics systems (Arvis et al., 2023; OECD, 2023; UNCTAD, 2023). It is appropriately applied where governance authority is distributed across multiple institutional actors with no single comprehensive authority, where infrastructure investment is advancing ahead of the governance architecture required to protect it, and where adequate secondary data and structured expert testimony are available to support evidence-coded SQCA assignment—conditions met by the three corridors assessed here and broadly characteristic of logistics governance environments across South Asia, Southeast Asia, Sub-Saharan Africa, and Latin America. It should not be applied in advanced industrial logistics economies where governance architecture has reached institutional maturity across all three pillars (the LPI remains the more appropriate operational benchmark there); in purely node-level assessment contexts such as individual port performance,

where the UNCTAD Port Performance Scorecard operates at a more appropriate resolution; or where the research question concerns operational efficiency, trade facilitation, or maritime connectivity rather than governance architecture.

11.2.2 Accordingly, Sustainability (S^1), Sovereignty (S^2), and Shock-Resilience (S^3) are treated as jointly necessary governance conditions whose absence cannot be offset by strengths elsewhere (Ferdows & De Meyer, 1990; Hohenstein et al., 2015) (and could be relaxed where the research question specifically concerns substitutability); that sustainability, sovereignty, and shock-resilience are jointly necessary conditions under the 2020s' compound disruption environment specifically, and that the pillar architecture should be revisited as context evolves; that the minimum five-source data requirement assumes institutional data accessibility; and that the expert-validation protocol assumes availability of practitioners with substantial direct corridor-governance experience.

11.2.3 Generalisability claims are correspondingly bounded, consistent with the study's critical realist stance: statistical representativeness is not claimed, only theoretical transferability; causal inference is not claimed from the reported association; and cross-national direct equivalence is not claimed without recalibration against the institutional documentation norms of other national contexts. The framework operationalises these governance conditions through a non-compensatory minimum-condition rule derived from Qualitative Comparative Analysis, whereby the weakest governance pillar determines the composite governance classification because the absence of a necessary institutional capability constrains overall governance maturity irrespective of strengths in other dimensions (Ragin, 2008; Schneider & Wagemann, 2012). The appropriate extension pathway is iterative, involving the application of S^3 -LOGIC across $n \geq 10$ corridors in at least three distinct emerging-economy contexts to provide an evidence base for inferential testing at an appropriate scale.

12. Conclusion

12.1 India's logistics transformation is a significant infrastructure modernisation programme and it is advancing ahead of the governance architecture required to protect its gains under disruption, maintain institutional control over its operational data and digital platforms, and ensure its sustainability trajectory is not eroded by demand-induced rebound. This asymmetry is not primarily a technical or commercial problem; it is a governance-architecture problem, and making it visible requires a governance diagnostic instrument. While the Digital Personal Data Protection Act, 2023 establishes India's general legal framework for personal data governance the findings suggest that sector-specific institutional arrangements remain necessary to govern operational logistics data, platform accountability, and sovereign decision-making and the proposed National Logistics Data Sovereignty Act is conceived as a complementary, sector-specific framework that extends rather than replaces the existing legal architecture.

12.2 The present study also establishes a foundation for future research. While the framework has been demonstrated through three strategically distinct freight corridor archetypes using a qualitative comparative approach, broader empirical validation across additional logistics corridors, ports, inland waterways, airports, and international transport systems would further strengthen its analytical robustness and external validity. Future studies may incorporate longitudinal datasets, quantitative weighting techniques, machine-learning-assisted governance diagnostics, and cross-country comparative analyses to refine the framework and evaluate its predictive capability under different institutional and geopolitical conditions. Beyond logistics, the conceptual architecture of Sustainability, Sovereignty, and Shock-Resilience may also prove valuable for assessing governance maturity in other critical infrastructure sectors, including energy, digital infrastructure, healthcare, water security, and smart-city ecosystems. Accordingly, S³-LOGIC should be viewed not as a static assessment model, but as an evolving governance architecture capable of adapting alongside the increasingly complex systems it is intended to evaluate.

12.3 The S³-LOGIC framework provides such an instrument. Applied across three structurally differentiated Indian freight corridor archetypes, it generates governance-vulnerability profiles that are internally consistent, practitioner-validated, and directly relevant to policy reform, demonstrating that governance fragility consistently originates in sovereignty and shock-resilience deficiencies rather than infrastructure availability alone-supporting the governance-first proposition advanced throughout this study, and consistent with the institutional emphasis of the OECD Recommendation on the Governance of Infrastructure (2020)([OECD, 2023](#))

12.4 These findings' strategic and geopolitical dimensions extend beyond operational logistics management. As India's freight corridors assume greater strategic importance through initiatives such as the India–Middle East–Europe Economic Corridor (IMEC), institutional resilience increasingly depends on coordinated governance, digital sovereignty, and adaptive disruption-response capability. S³-LOGIC provides the diagnostic; the UN Decade of Sustainable Transport provides the mandate. Accordingly, the framework should be interpreted as a governance-diagnostic complement to existing infrastructure and logistics performance benchmarks, enabling policymakers to identify institutional constraints before they manifest as operational vulnerabilities.

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